

SOHAIL MULLA

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SUMMARY

Data Scientist building ML applications from research to production. Skilled in Python, SQL, deep learning (PyTorch), and modern AI (GPT-4, RAG, embeddings). Strong foundation in experimentation, statistical modeling, and cloud deployment. 2+ years driving data-driven decisions in FinTech.

SKILLS

Languages: Python, R, SQL, C++, JavaScript

ML/AI Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, BERT, GPT-4, Sentence Transformers, Pandas, NumPy, NLTK, OpenCV

Techniques: A/B Testing, Causal Inference, Regression Analysis, Time Series Forecasting, NLP, Deep Learning, RAG, Feature Engineering, Statistical Modeling

Tools & Platforms: AWS, Azure, Docker, Git, Airflow, FastAPI, Flask, Streamlit, Jupyter, Vector Databases, Tableau, Superset, QlikSense

Analytics: Cohort Analysis, Funnel Optimization, User Segmentation, Product Analytics, EDA

EDUCATION

Syracuse University, School of Information Studies, Syracuse, NY

August 2024 - May 2026

M.S.| Applied Data Science

GPA: 3.92 / 4.00

- Coursework: Agentic AI, Database Administration, Applied Machine Learning, Natural Language Processing, Deep Learning,

Birla Institute of Technology and Sciences Pilani, Goa

August 2018 - July 2022

B.E. Honors| Mechanical Engineering

GPA: 3.12 / 4.00

- Coursework: Particle Image Velocimetry (PIV), Heat Transfer, Microfluidics, Computational Fluid Dynamics.

EXPERIENCE

ML & AI Integration Engineer, Syracuse University, Syracuse, New York

January 2026 -Present

- Developing Python MCP server integrating Claude AI with building simulation platform using function calling with specialized tools for automated parameter configuration from natural language queries
- Training neural networks on 1000+ parametric simulation runs to create surrogate models predicting energy/comfort metrics from building parameters, reducing simulation time from hours to seconds via ML inference.
- Building GUI components (6-module Hub, Refiner tool) via Python and Rhino/Grasshopper API for configuring energy, airflow, and IAQ simulations.

Computational Data Scientist, Syracuse University, Syracuse, New York

August 2025 - December 2025

- Developed Python bindings for OS-MOSIS, an open-source GPU solver built in C++ and CUDA, to enable seamless integration with Python for fluids and data science community.
- Implemented Fortran-style memory layout for MATLAB compatibility and C++ template dispatch for flexible precision (float32/float64); validated via Taylor vortex simulations with less than 0.05% error
- Packaged as a pip-installable library; expanding solver access from MATLAB only to broader Python scientific community.

Business Analyst, Freo, Bangalore, India

June 2023 - July 2024

- Devised targeted marketing strategies by identifying specific customer cohorts within diverse gold loan segments, leveraging Python, SQL, and Superset for customer data analysis and segmentation, resulting in a 40% increase in monthly revenue from gold loan revenue.
- Designed dynamic dashboards in Superset across sales and customer success operations for Fit Credit and Freopay insurance products, improving sales efficiency 15% through real-time insights and reducing CS manual work 50% (40 to 20 hrs./week) via intelligent ticket prioritization.
- Pre-processed data and applied NLP techniques with Python, pandas, NLTK, and scikit-learn to construct a sentiment analysis model, helping the Customer Success team identify sentiments in support tickets.
- Optimized 7-8 critical business KPI reports by migrating from a legacy database to Amazon Redshift, rewriting queries, and creating new tables to improve performance. Optimized query run time for these reports by 50%, cutting execution time from 50 minutes to 20 minutes, boosting Airflow workflow efficiency and eliminating report bottlenecks.

Product Analyst, Simpl, Bangalore, India

July 2022 - May 2023

- Designed and optimized a Customer Acquisition Cost (CAC) framework, achieving a 30% reduction in CAC for Simpl, utilizing SQL, Statistical Modeling and Excel for calculation and optimization.
- Created and maintained 12 QlikSense dashboards to track KPIs and business performance; collaborated cross-functionally to build automated data pipelines using GitHub and Airflow, ensuring data accuracy while reducing manual reporting by 85%.
- Conducted A/B testing on product launches and provided recommendations to increase user activation by 5% and network activation by 9%.

PROJECTS

GitHub Repository Assistant

- Developed an AI-powered GitHub repository analysis system using LangChain agents with ReAct framework to answer natural language queries about repository history, pull requests, and codebase evolution
- Engineered a semantic search pipeline integrating OpenAI embeddings (text-embedding-3-small) with ChromaDB vector database to index and retrieve relevant commits, PRs, and code snippets from 100+ commits in under 2 minutes.
- Implemented RAG (Retrieval-Augmented Generation) architecture with metadata filtering and document chunking strategies to provide contextually accurate answers about software development history and code implementations.

Smart Gym Cloud Platform

- Developed full-stack application on Azure with FastAPI backend (13 REST endpoints, SQL Alchemy ORM), React frontend, and serverless billing automation managing 300 members and 500+ registrations
- Designed AI recommendation engine with OpenAI function calling with 5 custom tools (profile analysis, class filtering, schedule validation, match scoring, collaborative filtering) generating personalized workout plans
- Deployed across Azure Static Web Apps (GitHub Actions CI/CD), Linux VM (Ubuntu), Azure Functions (timer-triggered), with Network Security Groups, CORS configuration, and API proxy architecture